

# ign1t10n: macOS Installer Specification

One install: a multi-validator F1R3Node-Rust shard, its observer, the F1R3Gaze browser, and the F1R3Games portal and games, served to any web browser

Version 0.4 (draft for review)

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## Abstract

This specification defines ign1t10n, an installer for macOS that leaves a person, a few minutes after double-clicking one file, with a working F1R3Node-Rust shard on their own machine (a bootstrap node, two to ten validators, and a read-only observer), the F1R3Gaze browser configured to deploy to that shard and read from it, and, new in this version, the F1R3Games portal and its games, installed on the shard and served on `http://localhost` so that an ordinary web browser (Safari, Chrome, Firefox) can sign on, launch new game instances, play them, and browse each game's gallery of plays. The portal is `f1r3games-service`, the Rust service of the F1R3Games repository, supervised by ign1t10n as one more process beside the nodes and Embers; each game client is served from an origin of its own, as the portal's frame model requires. Installation of the games on the chain (the portal's environment, each game's environment, and the registration of the games by a local stand-in for the F1R3FLY.io Cooperative) is automated, resumable and recorded in the manifest, replacing the eleven manual steps the first live installs of 5 and 6 October needed.

The specification is grounded in ign1t10n main at 46faa56 (the first working implementation of version 0.3), F1R3Games at 57a3b27 (branch `f1r3beat`, the portal with F1R3Pix and F1R3Beat), F1R3Node-Rust at `fce422a`, F1R3Gaze at `f6ee26a`, and the F1R3Games portal, F1R3Pix and F1R3Beat design documents.

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# 1 What changed in version 0.4

1. **F1R3Games is a component of the install** (default on, like Embers): the portal service, the F1R3Pix and F1R3Beat web clients, the games' environments on the shard, their registration, and a "F1R3Games" item in the menu bar that opens the portal in the person's default browser. New: Findings 11–21, Principles 1–3, §10 (the component), stages G1–G7 (§10.4), work packages F1–F7 in the F1R3Games repository (§12.3) and H7–H9, R2, T2 here, acceptance tests A15–A23, and decisions 9–17.
2. **The decisions of 30 September are folded in** (§18.2): the separate bootstrap is kept; Apple silicon only; macOS 13.0 minimum; 10 000 000 F1R3 to the browser's wallet; deploys go to a randomly selected active validator; timings tuned for a laptop; the node refuses HTTP requests carrying a foreign Origin (N5); Embers is an installation option, default on.
3. **What the first working build taught is adopted as specification** (from docs/IMPLEMENTATION.md at 46faa56): the supervisor is a classic per-user launch agent loaded with `launchctl bootstrap`, not an `SMAppService` agent; the genesis ceremony waits for  $N_0 - 1$  approvals; all six ports and the network id go on each node's command line; the bootstrap's node id is read from its output; node logs go to standard output; templates are compiled into the binary; the archive is `macos-arm64`.

## 2 Scope and conventions

### 2.1 What the installer delivers

After a successful install on a supported Mac, with no further action beyond approving a background item in System Settings when macOS asks:

1. `/Applications/F1R3Gaze.app` and `/Applications/ign1t10n.app` are installed, signed and notarised.
2. A per-user launch agent, `io.f1r3fly.ign1t10n.supervisor`, is loaded and running. It supervises:
  - a *bootstrap* node, which coordinated the genesis ceremony and is every other node's rendezvous; it is not bonded and never proposes;
  - $N$  *validators*, bonded at genesis or by a later resize, which propose and finalise blocks ( $N = 2$  unless the person chose otherwise,  $2 \leq N \leq 10$ );
  - one *observer*, a read-only node that serves every read;
  - Embers, unless the person turned it off;
  - the *portal*, `f1r3games-service`, unless the person turned F1R3Games off.
3. Every key was generated on this machine at install time. The shard's genesis funds the person's active F1R3Gaze wallet and a local faucet; the faucet funds Embers, the portal and the local Cooperative.
4. F1R3Gaze deploys to a randomly selected active validator, reads from the observer, and is open in a window.
5. The F1R3Games portal environment and the five game environments are installed on the shard; F1R3Pix and F1R3Beat are registered; the portal answers at `http://localhost:40700` and has been opened in the default browser, where the person can create a keystore, be funded by the faucet, launch a game, and browse its gallery.
6. Every listening socket of every supervised process is bound to a loopback address.

7. The shard and the portal survive logout, sleep and reboot, and come back with the chain, the portal’s origin and every browser keystore intact, at the size last chosen. Throughout, “*N nodes*” in the user interface means *N* validators; the bootstrap and the observer are always present and are shown separately. With the defaults, five `f1r3node` processes, Embers and the portal run.

## 2.2 Non-goals for this version

More than ten validators; joining a remote or public shard; exposing the local shard or the portal to the network (so invitations work for other browsers and other accounts on this Mac only); stake other than equal stakes; Intel Macs, Linux and Windows; Docker; running the `F1R3Games` portal *inside* `F1R3Gaze` (it executes no JavaScript; §10.9); web clients for `F1R3Ink`, `F1R3SideChat` and `F1R3Skein`, which do not exist yet; the `F1R3Skein` visionOS instrument.

## 2.3 Pinned baselines

| Source                     | Revision                      | Notes                                                                                                                                                                                                                                                                                                                                           |
|----------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ign1t10n</code>      | <code>main @ 46faa56</code>   | “first working version” of v0.3; <code>docs/IMPLEMENTATION.md</code> records its decisions, deviations and known issues                                                                                                                                                                                                                         |
| <code>F1R3Node-Rust</code> | <code>master @ fce422a</code> | <code>crate node v0.4.46</code> ; <code>toolchain nightly-2026-02-09</code> ; topology reference <code>docker/shard.yml</code>                                                                                                                                                                                                                  |
| <code>F1R3Gaze</code>      | <code>main @ f6ee26a</code>   | release v0.1.0; crates <code>gaze-shard</code> , <code>gaze-wallet</code> reused by <code>ign1t10n</code> ; renders HTML with Blitz and runs no JavaScript                                                                                                                                                                                      |
| Embers                     | <code>28e50d4</code>          | bundled as <code>Helpers/embers</code> ; its contracts do not parse on <code>fce422a</code> (known issue)                                                                                                                                                                                                                                       |
| <code>F1R3Games</code>     | <code>f1r3beat 57a3b27</code> | @ <code>Portal/f1r3games-portal</code> (Rust workspace: <code>core</code> , <code>wallet</code> , <code>node</code> , <code>service</code> , <code>cli</code> , <code>games</code> , <code>rhoint</code> ; React shell in <code>web/</code> ); <code>F1R3Pix/client</code> , <code>F1R3Beat/client</code> ; to be re-pinned to main once merged |
| Designs                    | 5–6 October 2026              | <code>F1R3Games</code> Portal design v0.1; <code>F1R3Pix</code> design v1; <code>F1R3Beat</code> design v2                                                                                                                                                                                                                                      |

## 2.4 Conventions

MUST, MUST NOT, SHOULD, SHOULD NOT and MAY carry their RFC 2119 meanings. “The state directory” is `~/Library/Application Support/F1R3FLY/ign1t10n`. “The profile” is `F1R3Gaze`’s profile directory. “The portal” is the running `f1r3games-service` process together with the web shell it serves; “the portal environment” is the games environment it installs on the shard; “a game environment” is one game’s contract. “The local Cooperative” is a key `ign1t10n` generates to stand in for the `F1R3FLY.io` Cooperative on this shard (Decision 10). “An origin” is a scheme, host and port, as browsers define it. Ports are written for the default allocation.

## 3 Findings

These are the facts the design rests on. Findings 1–10 are carried from version 0.3, shortened where the implementation has since settled them; Findings 11–21 are new.

### 3.1 The shard

**Finding 1** (The observer is not optional). Every endpoint that runs an exploratory deploy is served only by read-only nodes; a validator answers 400 `readonly_node_required`. That covers `POST /api/explore-deploy`, `POST /api/estimate-cost`, `GET /api/registry/{uri}` and `GET /api/balance/{address}` (`docs/node/api-reference.md`). `F1R3Gaze`, `Embers` and `f1r3games-service` all read there.

**Finding 2** (The tested topology). `docker/shard.yml` runs a bootstrap started with `--ceremony-master-mode`, `--required-signatures` and `--heartbeat-disabled`, absent from `bonds.txt`; genesis validators started with `--genesis-validator` and `--bootstrap` pointing at it; and an observer bootstrapping from it. `ign1t10n` runs that shape on loopback, generalised from three validators to  $N$ .

**Finding 3** (Same host, distinct ports). On one Mac every node advertises `127.0.0.1` and differs by port. The node's port options have default values that override the configuration file, so all six ports go on the command line; the outgoing network id defaults to `testnet` unless `--network-id` is given (both found on the first real multi-node run).

**Finding 4** (Approvals). The node requires strictly fewer ceremony approvals than genesis validators; `shard.yml` uses 2 of 3. `ign1t10n` passes  $\max(N_0 - 1, 1)$ .

**Finding 5** (The shard can be resized while running). A node that was not at genesis joins by last-finalised-state fetch, is funded, submits a bond deploy to a *bonded* validator, and becomes active at the next epoch boundary (`docs/node/joining-a-network.md`); leaving is `PoS!("withdraw", ...)`, effective at the next boundary, with payout after the quarantine. `epoch-length` and `quarantine-length` are 10, so both take seconds to minutes locally. A bonded validator that does not propose is counted in consensus while contributing nothing.

**Finding 6** (Deploys can be signed outside the node). `F1R3Gaze`'s `gaze-shard` produces deploys byte for byte as the node verifies them; `gaze-wallet` renders the vault transfer. `ign1t10n` uses both for faucet transfers, bond, withdraw and sweep deploys.

**Finding 7** (The shipped development keys are public). `docker/.env.example` and the committed `docker/.env` carry the private keys of every node in the Docker topologies and control the genesis supply there. The `F1R3Games` repository likewise commits example keys under `Portal/f1r3games-portal/examples/local-shard/` (a service key, an environment key, a token secret and five game environment keys). No install may reuse any of them.

**Finding 8** (How a key can reach a process). `--validator-private-key` exposes the key in the argument vector; `--validator-private-key-path` takes an encrypted PEM whose password comes from `F1R3NODE_VALIDATOR_PASSWORD`. `f1r3games-service` reads its keys only from plaintext key files named in its configuration (`service_key_file`, `env_key_file`, `token_secret_file`) and the game keys from a directory (`--keys`); it reads them once, at start.

**Finding 9** (Bind addresses and origins). `api-server.host` is configurable; the protocol transport and discovery bind `0.0.0.0` until `N1` lands. With `N5`, the node answers 403 to any HTTP request carrying an `Origin` header, so no page in a stock browser can use a local node directly; programs that send no `Origin` (`F1R3Gaze`, `Embers`, `f1r3games-service`) are unaffected.

**Finding 10** (Lifecycle facts). The node handles `SIGTERM`. A joined node restarted on the same data directory reconnects in about a second; its identity is the TLS certificate in the data directory, so a data directory must never be deleted while peers remember it. During the ceremony the bootstrap cannot answer `/api/status`; its node id is read from what it prints at start.

## 3.2 F1R3Games

**Finding 11** (The portal is one Rust service). `f1r3games-service` (`crate f1r3games-service, axum`) serves the API under `/api` and, as a fallback service, the built React shell from `static_dir`, on the single address in `listen` (default `127.0.0.1:8640`). It holds three secrets: the *service key*, which pays the environment deploys and the testnet faucet; the *environment key*, which signs the portal environment's `insertSigned` registration and so fixes its registry URI; and a *token secret*. It deploys to one `validator_url` and reads at one `observer_url`; it never proposes (validators propose by heartbeat, which `ign1t10n` leaves on). Subcommands: `serve` (default; installs the environment first when `bootstrap_env` is true), `bootstrap`, `env-uri`, `keygen`, `games-keygen`, `games-install --keys DIR --version V, games-manifests --keys DIR --entry-base URL -out FILE`. The portal design placed this logic in an Embers domain; the implementation chose a dedicated service, and the implementation governs here: F1R3Games does not need Embers.

**Finding 12** (The browser never reaches the node). The shell's wallet is Rust compiled to WebAssembly (`crates/wallet`); it signs in the page and hands signed deploys to `f1r3games-service (/api/prepare, /api/send)`; reads go through `/api/explore` and the GET routes. The page holds no node URL. N5 therefore stands without exception.

**Finding 13** (Games run on origins of their own). The portal frames a game in a sandboxed `iframe` (`allow-scripts allow-same-origin allow-forms allow-pointer-lock allow-downloads`) and answers its host protocol by `postMessage`, accepting only messages whose `event.origin` is the origin of the game's registered entry and whose source is that frame (`web/src/core/host.ts`). `allow-same-origin` is safe only because the game is on another origin, and `GameFrame.tsx` refuses to frame a game served from the portal's own origin. A game client is a static Vite build (`base: "."`) loaded as `<entry>?instance=<id>&portal=<origin>`; its gallery renderers are `<entry>preview/<kind>.html`. F1R3Pix (renderer `canvas`) and F1R3Beat (pattern, `session`) have clients; F1R3Ink and F1R3SideChat have only environments; F1R3Skein's manifest lists `visionos` and `web` and has no web client.

**Finding 14** (Browser state belongs to an origin). The keystore, contacts and the pinned environment URI live in IndexedDB under the portal's origin (`web/src/core/store.ts`). A passkey is created with no explicit relying-party id, so it is bound to the origin's host; WebAuthn refuses IP literals as relying-party ids, and `localhost` is a secure context in Safari, Chrome and Firefox. Hence the portal MUST be reached as `http://localhost:port`, and that origin MUST not change for the life of the install: at `127.0.0.1:port`, or at another port, the person meets an empty keystore and passkeys that do not work.

**Finding 15** (Keys fix the environment URIs, and wallets pin them). The portal environment's URI is derived from the environment key; each game environment's URI from that game's key; every call template in a manifest embeds its game's URI, so the manifest's template hashes depend on the keys. At sign-on the shell pins the portal environment URI and afterwards refuses a service that serves another (`web/src/core/portal.ts`). Rotating the environment key therefore strands every keystore in every browser.

**Finding 16** (Installation is an ordered sequence of deploys). On a new chain: fund the service key; install the portal environment (one deploy of up to `env_phlo_limit`, 50M phlo by default) and wait until the `env.probe` template reports the version; install each game environment (`games-install`); write the manifests (`games-manifests`), which take a single `--entry-base` and produce `<base>/<id>/`; register each with `games.register`, which the environment accepts only

from the `coop_address` that was rendered into it at install. Registration is today done with the `f1r3games` CLI, which needs its own passphrase-protected keystore holding the Cooperative's key (`register-games FILE [--only ID]`).

**Finding 17** (Raising an environment's version discards its state). Installing a higher `env_version` of the portal environment re-initialises its maps: profiles, instances, plays, invitations and sponsorships on that shard are not carried over (F1R3Games docs/IMPLEMENTATION.md). Replacing a game environment's body likewise replaces its per-instance state. A change to a game's call templates changes the manifest hashes and requires re-registration; a change confined to a web client changes nothing on chain.

**Finding 18** (The faucet pays from the service key). `POST /api/testnet/fund` transfers `faucet.amount dust` (default  $10^8$ , i.e. 1 F1R3) from the service key to an address the page names, when `faucet.enabled`; the shell calls it during sign-on so a new key can pay for its first deploy.

**Finding 19** (What the first live installs needed by hand). Standing the portal up on an `ign1t10n` shard on 5 and 6 October took: building the service and shell (including `npm install --include=dev` and `npm's install-script` approval for `esbuild`); `keygen` and `games-keygen`; reading the shard's endpoints with `ctl env`; funding the service key and a stand-in Cooperative with `ctl fund`; choosing a stand-in Cooperative address; editing `f1r3games.toml`; `bootstrap` and waiting; `games-install`; `games-manifests`; importing the Cooperative key into the CLI and `register-games`; and serving the game clients from a second development server (`:5174`) so they had their own origin. Defects found only on the live shard (the `deploy-id` parse, multi-line templates, `validAfterBlockNumber` from the validator's sequence number, `Nil` dropped from tuples, escaped manifest sources, `allow-same-origin`, `.contains` on lists, a `/\ List` binder) are fixed at the pinned revision. Every manual step above becomes a stage of `ign1t10n` (§10.4); no developer tool is needed on the person's Mac.

**Finding 20** (`localhost` has two addresses). `localhost` resolves to `::1` and `127.0.0.1`, and browsers try both. A service listening on one family leaves the other free for any program to bind, and that program then answers some of the browser's requests: the same hazard version 0.3's implementation met with Docker's `*:p` listeners (`ports::port_free`).

**Finding 21** (F1R3Gaze runs no JavaScript). F1R3Gaze renders HTML and CSS with Blitz and executes no JavaScript; behaviour is `f1r3lang`. The React shell therefore runs only in a stock browser. The shell is cut so that a later `f1r3lang` rendition can replace it (docs/GAZE-MAPPING.md); that needs capabilities F1R3Gaze lacks today (allowances, `invite`, `open`, `pay`, `frame`, per-site storage).

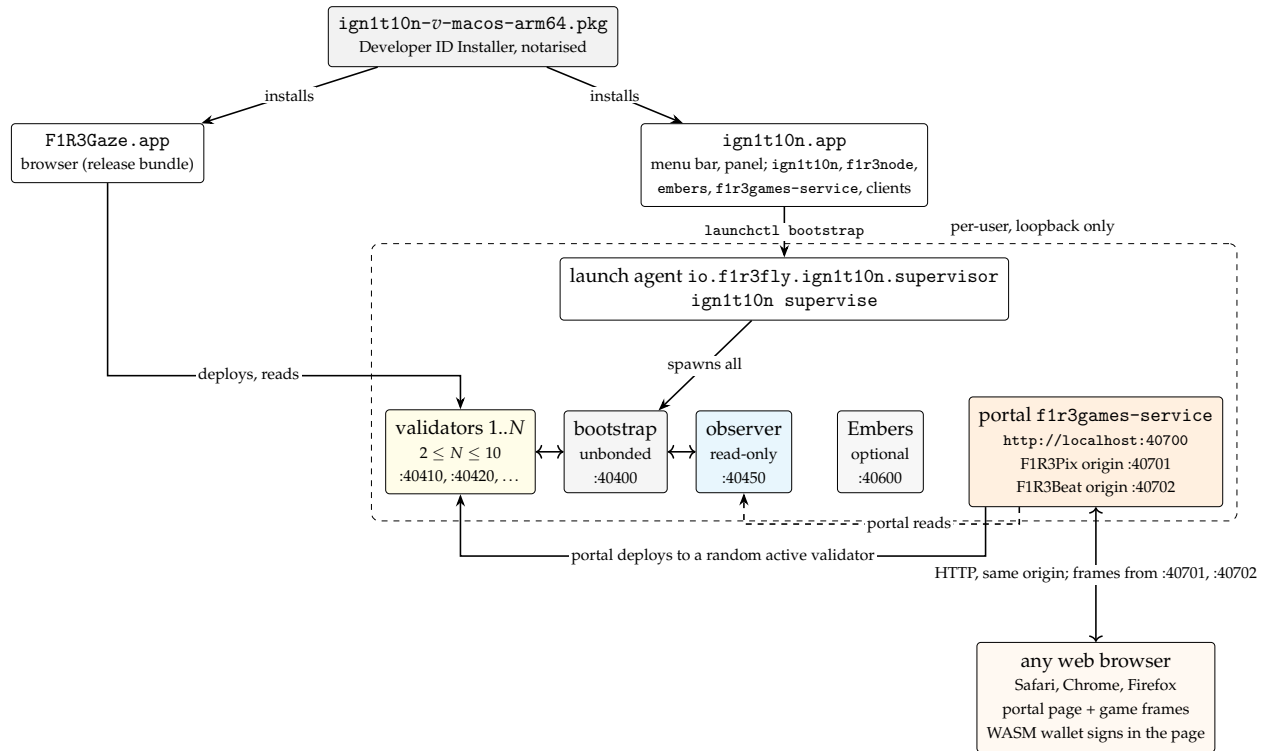


Figure 1: What the installer lays down and what runs afterwards, with the defaults. Nodes speak the peer protocol to one another through the bootstrap. The browser reaches only the portal; the portal alone reaches the nodes.

## 4 Architecture

### 4.1 Components

| Component                       | Role                                                                                                                                                                                                                                                                            |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F1R3Gaze.app                    | The F1R3Gaze release bundle, byte for byte.                                                                                                                                                                                                                                     |
| ign1t10n.app                    | Menu-bar application and configuration panel; provisions, starts, stops, resizes, resets and reports on the shard and its services. Contains ign1t10n (one binary, three roles), f1r3node, embers, f1r3games-service, the f1r3games CLI, the portal shell and the game clients. |
| launch agent                    | io.f1r3fly.ign1t10n.supervisor; runs ign1t10n supervise, which owns every process below.                                                                                                                                                                                        |
| bootstrap, validators, observer | f1r3node run, as in version 0.3 (§8.1).                                                                                                                                                                                                                                         |
| Embers                          | Optional; deploys to one active validator, reads at the observer.                                                                                                                                                                                                               |
| portal                          | f1r3games-service serve: the F1R3Games API and shell at <code>http://localhost:40700</code> , and each bundled game client at an origin of its own (F2); deploys to one active validator, reads at the observer.                                                                |
| F1R3Games jobs                  | One-shot runs of f1r3games-service for installation and registration (§10.4) and, if enabled, the F1R3Beat breeder (§10.7).                                                                                                                                                     |
| faucet                          | A key held by ign1t10n, never by a process, whose vault holds the undistributed genesis supply; funds joining validators, Embers, the portal, the local Cooperative and “Fund a wallet”.                                                                                        |
| local Cooperative               | A key held by ign1t10n; the address rendered into the portal environment as <code>coop-address</code> ; signs only <code>games.register</code> and <code>games.retire</code> .                                                                                                  |

### 4.2 Principles

Version 0.3’s principles stand: per-user and unprivileged at run time; loopback only; fresh secrets; one owner for the shard; a bonded validator runs; touch only what was created; idempotent, resumable operations. Version 0.4 adds three.

**Principle 1** (Stable origins). The portal’s origin, `http://localhost:port`, is fixed when it is first allocated and never changes automatically: not at start, not on a port conflict, not at reset, not at upgrade. Every browser keystore lives there (Finding 14). The same holds for each game’s origin, since manifests on the chain name it.

**Principle 2** (The browser reaches only the portal). No page served by ign1t10n is given a node URL, and no node accepts a browser’s request (N5). Everything a page does on the chain goes through `f1r3games-service`.

**Principle 3** (Keys outlive chains). The portal’s environment key and the games’ keys are kept across resets and upgrades, so the environment URIs, and with them every pinned keystore and every manifest hash, survive a new chain (Finding 15). They change only when the person explicitly asks to rotate them.

## 5 The distribution artifact

**Requirement 1** (Product archive). The release artifact **MUST** be a flat product archive, `ign1t10n-v-macos-arm64.pkg`, built with `productbuild` from two component packages

(`io.f1r3fly.f1r3gaze.pkg`, `io.f1r3fly.ign1t10n.pkg`), signed with a *Developer ID Installer* certificate, notarised and stapled. Each component MUST install its bundle to `/Applications` with bundle-version checking on, so an installed newer F1R3Gaze is not downgraded.

**Requirement 2** (The post-install hook does one thing). The `postinstall` script of the `ign1t10n` component package (`io.f1r3fly.ign1t10n.pkg`) MUST do nothing but open `ign1t10n` in the console user's session with `--first-run`, and MUST exit 0 even if that fails. It MUST NOT create files in the user's home, load agents or generate keys. (Known issue at 46faa56: nothing is visible between the installer's Close and `ign1t10n`'s window; `ign1t10n` SHOULD show its window before it restarts an existing supervisor.)

**Requirement 3** (Platforms). Apple silicon only; every Mach-O is arm64; minimum macOS 13.0 (resolved decisions 2 and 3).

**Requirement 4** (No dynamic dependencies outside the system). For every Mach-O in the archive, including `f1r3games-service` and `f1r3games`, `otool -L` MUST list only paths under `/usr/lib` and `/System/Library`.

**Requirement 5** (No developer tools at install time). The portal shell, its WebAssembly wallet and the game clients MUST be shipped built. Nothing on the person's Mac runs `npm`, `cargo`, `wasm-bindgen` or `Vite`.

## 6 ign1t10n

### 6.1 Bundle layout

|                                                              |                                                                                                                                  |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>ign1t10n.app/Contents/</code>                          |                                                                                                                                  |
| <code>Info.plist</code>                                      | <code>io.f1r3fly.ign1t10n</code> , <code>LSUIElement</code> , arm64, min 13.0                                                    |
| <code>MacOS/ign1t10n</code>                                  | menu bar (default), <code>`supervise`</code> , <code>`ctl`</code>                                                                |
| <code>Helpers/f1r3node</code>                                | <code>f1r3node-rust`node` v0.4.46</code>                                                                                         |
| <code>Helpers/embers</code>                                  | Embers 28e50d4                                                                                                                   |
| <code>Helpers/f1r3games-service</code>                       | the F1R3Games portal service (new)                                                                                               |
| <code>Helpers/f1r3games</code>                               | the F1R3Games CLI (Decision 16) (new)                                                                                            |
| <code>Resources/</code>                                      |                                                                                                                                  |
| <code>ign1t10n.icns</code> , <code>StatusTemplate.pdf</code> | F1R3FLY.io brand kit (firefly bug)                                                                                               |
| <code>compat.toml</code>                                     | storage formats this node can open                                                                                               |
| <code>versions.toml</code>                                   | what this release bundles (read at start)                                                                                        |
| <code>F1R3Gaze.app</code>                                    | installed by S0 if F1R3Gaze is missing                                                                                           |
| <code>f1r3games/</code>                                      | (new)                                                                                                                            |
| <code>games.toml</code>                                      | per game: id, env version, template hashes, client                                                                               |
| <code>present</code>                                         |                                                                                                                                  |
| <code>portal/</code>                                         | <code>web/dist</code> : the shell, fonts, the wallet <code>.wasm</code>                                                          |
| <code>games/f1r3pix/</code>                                  | <code>F1R3Pix/client/dist</code> ( <code>index.html</code> , <code>preview/canvas.html</code> )                                  |
| <code>games/f1r3beat/</code>                                 | <code>F1R3Beat/client/dist</code> ( <code>index.html</code> , <code>preview/pattern.html</code> ,<br><code>session.html</code> ) |
| <code>uninstall.sh</code>                                    |                                                                                                                                  |

The clients are served read-only from inside the signed bundle, so an upgrade replaces them atomically and nothing in the state directory can alter them.

## 6.2 One binary, three roles

`ign1t10n` (Rust 1.91) is the menu bar and panel when run with no arguments, the supervisor as `ign1t10n supervise`, and a client of the control socket as `ign1t10n ctl verb`. Version 0.4 adds the verbs `games on|off`, `games status [--json]`, `games reinstall`, `games open`, `games env` (prints the portal URL and the variables that point the `f1r3games` CLI at it) and `games breeder on|off`. The control socket is `run/control.sock`, one JSON object per line, peer uid checked.

## 6.3 The launch agent

A classic per-user agent, `~/Library/LaunchAgents/io.f1r3fly.ign1t10n.supervisor.plist`, naming the supervisor by absolute path and loaded with `launchctl bootstrap gui/<uid>`. `ign1t10n` rewrites and reloads it whenever the path or contents differ, and removes any `SMAppService` registration left by earlier builds. (Adopted from 46faa56: after a reinstall, `launchd` kept a stale background-task entry for the `SMAppService` agent and never started it again.) The agent still appears under Login Items. `KeepAlive` restarts only on an unsuccessful exit; `ExitTimeOut` covers the ordered shutdown.

## 6.4 The state directory

```
~/Library/Application Support/F1R3FLY/ign1t10n/      0700
shard.toml                      the manifest (Appendix B), schema 4
keys/ genesis/ conf/ nodes/ run/ archive/ as in version 0.3
games/                          (new)
  f1r3games.toml                the portal's configuration (Appendix A.2)
  manifests/<id>.json            each game's manifest as registered
  cli/                          F1R3GAMES_HOME for jobs that need the CLI
~/Library/Logs/F1R3FLY/ign1t10n/
ign1t10n.log, <node>.stdout.log, embers.stdout.log
portal.stdout.log, games-job.log                          (new)
```

## 6.5 Keys and Keychain items

Node and faucet keys are as in version 0.3: encrypted PKCS#8 PEMs, each with its own random password in a Keychain item of service `io.f1r3fly.ign1t10n`, account `pem.name`; Embers's secrets in `embers.secrets`. `F1R3Games` adds one item, `games.secrets`, a JSON object holding the service key, the environment key, the token secret, one environment key per game, the local Cooperative's key and the breeder's key. Its access control list is `ign1t10n`'s designated requirement. A key reaches a `F1R3Games` process only in that process's own environment, and only the keys the process needs (§10.3).

# 7 Provisioning

Provisioning runs in the menu-bar process, as the user, when `shard.toml` is absent or records an incomplete stage. Stages S0–S9 are version 0.3's, as implemented; the first-run window's "Advanced" disclosure now offers the validator count (default 2), "Bundle Embers" (default on), "Install F1R3Games" (default on) and "Open F1R3Games when ready" (default on).

**S0 preflight**

macOS  $\geq 13$ ; Apple silicon; F1R3Gaze present (else install the bundled copy); memory and disk for  $N_0 + 2$  nodes plus Embers and the portal (§7.1); the state directory creatable.

**S1 ports**

node blocks of six as in version 0.3 (bootstrap 40400, validators 40410–40440 then 40460–40510, observer 40450), Embers 40600, and, when F1R3Games is chosen, the portal and game ports of §10.2. A port is free only if it binds *and* nothing answers a connection to it, on 127.0.0.1 and, for the portal and games, on  $::1$ .

**S2 the paying wallet**

F1R3Gaze’s active wallet, or a new one labelled “Local shard”.

**S3 keys**

bootstrap, validators  $1..N_0$ , faucet; Embers secrets if chosen; `games.secrets` if chosen and absent (stage G1).

**S4 genesis files**

`bonds.txt` with equal stakes of 1 000; `wallets.txt`: 10 000 000 F1R3 to the browser’s wallet, 1 000 F1R3 to each validator, the rest of a 500 000 000 F1R3 supply to the faucet.

**S5 node configurations**

`common.conf` and one file per node (Appendix A.1).

**S6 load the agent**

§6.

**S7 genesis**

bootstrap, then validators with `--genesis-validator`, then the observer; record the genesis hash, node version and storage format.

**S8 configure F1R3Gaze**

the managed block or `settings.d` drop-in, with a randomly selected active validator.

**S9 open F1R3Gaze**

after the observer is ready.

When F1R3Games is chosen, the supervisor continues with stages G1–G7 (§10.4) once the shard is Running, and the first-run window lists them below S9. A failure in G1–G7 leaves the shard Running and F1R3Games Failed; it never undoes S0–S9.

**7.1 Budgets**

Targets on an M-series Mac with  $N_0 = 2$ : S0–S6 under 10 s of machine time; genesis under 120 s; observer join under 180 s; G1–G7 under 180 s (six environment deploys and two registrations, each waited to finalisation, about one block each); whole first run under seven minutes.  $M_{\min}$  and  $D_{\min}$  add the measured footprint of Embers and of the portal (a placeholder of 150 MB each until T1 measures it). Plays are stored on chain, so disk grows with play: T2 measures the growth per published F1R3Pix canvas and F1R3Beat session across the  $N + 2$  data directories, and the panel shows it.

## 8 Supervision

### 8.1 Start

1. Read `shard.toml`; refuse to start if the node's storage format is not in `compat.toml`; probe every recorded port; on conflict go to §8.5.
2. Start the bootstrap; read its node id from its output.
3. Start every validator whose recorded state runs, in parallel; wait for each `/api/ready`.
4. Start the observer; wait for `/api/ready`.
5. If Embers is on: fund it once from the faucet; draw a random active validator; start it; wait for `/api/service/ready`.
6. Redraw F1R3Gaze's validator. Report Running.
7. If F1R3Games is on: run any incomplete G-stage or upgrade step (§10.8); draw a random active validator for the portal; start the portal; wait for readiness (§10.5). Report F1R3Games Running.
8. Resume an interrupted resize.

Node argument vectors (Appendix A.1 has the files they name):

```
# every node
f1r3node run --config-file "$STATE/conf/<node>.conf" \
  --protocol-port B --discovery-port B+4 --api-port-grpc-external B+1 \
  --api-port-grpc-internal B+2 --api-port-http B+3 --api-port-admin-http B+5 \
  --network-id "$NETWORK_ID" --host 127.0.0.1 --no-upnp --allow-private-addresses \
  --native-token-name F1R3CAP --native-token-symbol F1R3 --native-token-decimals 8
# bootstrap adds
--validator-private-key-path "$STATE/keys/bootstrap.pem" \
--required-signatures $(max(NO-1,1)) --ceremony-master-mode --heartbeat-disabled
# validator k adds
--bootstrap "rnode://$BOOT_ID@127.0.0.1?protocol=40400&discovery=40404" \
--validator-private-key-path "$STATE/keys/validator-k.pem" [--genesis-validator]
# observer adds
--bootstrap "rnode://$BOOT_ID@127.0.0.1?protocol=40400&discovery=40404" --heartbeat-disabled
```

The portal runs as `f1r3games-service -c "$STATE/games/f1r3games.toml" serve` with the reduced environment (`HOME`, `PATH`, `TMPDIR`, `LANG`) plus its keys (§10.3) and `RUST_LOG=info`; output goes to `portal.stdout.log`.

### 8.2 Health

Every 10 s the supervisor reads `/api/status` from every node, Embers's readiness and the portal's (§10.5). The shard's status machine is unchanged: `Stopped`  $\rightarrow$  `Starting(stage)`  $\rightarrow$  `Running`  $\rightleftharpoons$  `Degraded(reason)`, plus `Resizing(step)` and `Failed(reason)`. F1R3Games has its own: `Off`, `Installing(stage)`, `Running`, `Degraded(reason)`, `Failed(reason)`, `Waiting` (the shard is not Running). The menu shows both.

### 8.3 Restarts

A node that exits unexpectedly is restarted with exponential back-off (1, 2, 4, ..., 60 s); five unexpected exits of one node in five minutes stop the whole shard in `Failed` rather than leave a bonded

validator down. Embers and the portal are restarted the same way, but five exits in five minutes put only that service in Failed; the shard keeps running.

## 8.4 Stop

On SIGTERM: the portal (and any F1R3Games job) and Embers, waiting up to 30 s; the observer, 30 s; every validator at once, 60 s; the bootstrap, 30 s; SIGKILL whatever remains. “Stop shard” stops everything but the supervisor; uninstall and Shutdown also make it exit 0.

## 8.5 Port conflicts at start

If a node’s or Embers’s recorded port is taken, that component is Failed, naming the port and its holder (lsOf), and the menu offers “Choose new ports” for every clashing node at once. The portal and game ports are *not* offered a silent move (Principle 1): F1R3Games goes Failed naming the holder and asks the person to stop it. “Move F1R3Games to a new address...” exists in the panel behind a dialog that says browser keystores at the old address will not follow, offers to open the portal’s Wallet page so each key can be exported first, and then re-registers the games under their new origins (G6).

## 8.6 Sleep and wake

The processes run under a launch agent, so App Nap does not apply. The supervisor holds a PreventUserIdleSystemSleep assertion only during provisioning, resizes and G-stages.

# 9 Resizing the shard

Version 0.3’s procedure stands: slots grow (keyed, funded, joining, bonding, active) one after another and shrink (retiring, left, paid, absent) from the highest; the node is stopped only after it has left the active set; a failed step stops at that slot and offers Retry or Undo; “Start a new shard with *M* validators” is a reset. One addition: before a validator withdraws, *every* client pointed at it is moved to another active validator: F1R3Gaze’s setting, Embers, and now the portal, which is restarted with the new validator\_url (a restart costs about a second; deploys a page sent in that second fail and are retried by the page). With F7 the portal takes the whole list and needs no restart.

# 10 F1R3Games

## 10.1 What runs

| Process      | Command                                 | Purpose                                                                                                                        |
|--------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| portal       | f1r3games-service<br>serve              | API and shell at http://localhost:40700; F1R3Pix at http://localhost:40701, F1R3Beat at http://localhost:40702 (F2); faucet on |
| install job  | bootstrap;<br>games-install             | G4, G5; upgrades                                                                                                               |
| register job | games-manifests;<br>register-games (F3) | G6; re-registration                                                                                                            |
| breeder job  | beat-epoch (F6)                         | optional, daily (§10.7)                                                                                                        |

Only the portal is long-lived. Jobs are run by the supervisor one at a time, with the portal’s configuration, logged to `games-job.log`, and recorded in `shard.toml` before and after, so an interrupted job is re-run (every job is idempotent: each probes the chain before it deploys, Finding 16).

## 10.2 Origins and ports

| Origin               | Port        | Serves          | Manifest entry                   |
|----------------------|-------------|-----------------|----------------------------------|
| portal               | 40700       | /api, the shell | —                                |
| F1R3Pix              | 40701       | /f1r3pix/       | http://localhost:40701/f1r3pix/  |
| F1R3Beat             | 40702       | /f1r3beat/      | http://localhost:40702/f1r3beat/ |
| Ink, SideChat, Skein | 40703–40705 | reserved        | not yet registered               |

**Requirement 6** (The host is `localhost`). Every URL `ign1t10n` writes for the portal or a game (manifest entries, `portal_base_url`, the menu’s “Open F1R3Games”, “Copy F1R3Games address”) MUST use the host `localhost`. The portal MUST redirect a request whose Host is `127.0.0.1:p` or `::1:p` to `http://localhost:p` (308) and MUST refuse any other Host (421), which also defeats DNS rebinding (F4).

**Requirement 7** (Both loopback families). The portal and every game origin MUST listen on `127.0.0.1` and on `::1` at the same port (F4); a port is allocated only if it is free on both (Finding 20). Nothing listens on any other address.

**Requirement 8** (One origin per game). Each bundled game MUST be served from its own port, serving only its own directory, with `Content-Security-Policy: frame-ancestors http://localhost:40700` and `X-Content-Type-Options: nosniff` (Decision 12). Games then share neither storage nor a `postMessage` origin.

Ports are allocated once, at G2, from the plan or else the first free decade in 41700–49990, and recorded; they are never moved except by the person (§8.5). The block 40700–40709 is outside the ports a developer’s F1R3Games checkout uses (8640, 5173, 5174), so a development portal and the installed one run side by side against the same shard (Decision 9).

## 10.3 Keys

| Key               | Funded with    | Given to                                                                                                                          |
|-------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| service key       | 1 000 000 F1R3 | portal (F1R3GAMES_SERVICE_KEY); install job; register job (pays)                                                                  |
| environment key   | —              | portal (F1R3GAMES_ENV_KEY); install job                                                                                           |
| token secret      | —              | portal (F1R3GAMES_TOKEN_SECRET)                                                                                                   |
| game keys (5)     | —              | install job and register job only (F1R3GAMES_GAME_KEY_<ID>); F1R3Beat’s also to the breeder job once, for <code>setBreeder</code> |
| local Cooperative | 1 000 F1R3     | register job only (F1R3GAMES_COOP_KEY)                                                                                            |
| breeder           | 1 000 F1R3     | breeder job only (F1R3GAMES_BREEDER_KEY)                                                                                          |

All six kinds live in `games.secrets` and reach a process only in its environment (F1). Until F1 lands, `ign1t10n` writes the key files a command needs into a fresh 0700 directory under `run/`, passes their paths, waits until the process reports ready (the service reads its keys once, at start), and deletes the directory; the paths never appear in an argument vector except as file names.

## 10.4 Installation on the chain

Each stage is recorded in `shard.toml` under `[games.stages]` and resumes at its first incomplete step.

### G1 secrets

Generate `games.secrets` if it is absent; otherwise keep it (Principle 3). Record the service, Cooperative and breeder addresses and every environment URI (`f1r3games-service env-uri` and the game-key derivation).

### G2 ports

Allocate the portal's and the bundled games' ports (§10.2) unless already recorded.

### G3 fund

Faucet transfers to the service key, the local Cooperative and (if the breeder is on) the breeder; wait until each is finalised at the observer. Skipped for an address whose balance already covers its amount.

### G4 configure and install the portal environment

Render `games/f1r3games.toml` (Appendix A.2) with a randomly selected active validator; run the install job's bootstrap; wait until the `env.probe` template at the observer reports `env-version` (F5's status `--json` reports it directly).

### G5 install the game environments

`games-install --version V` with the versions from `Resources/f1r3games/games.toml`; wait until each game's probe reports its version. All five environments are installed, so registering a game whose client arrives later needs no deploy but the registration.

### G6 register

For each game whose client is bundled: render its manifest with its own origin as entry (F3, or one `games-manifests` run per game until F3); compare with `manifests/<id>.json` and with `GET /api/games/<id>`; if the game is absent or its template hashes or entry differ, sign `games.register` with the local Cooperative's key and wait until `/api/games` lists it with the same hashes. Save the manifest.

### G7 open

Start the portal (§8.1); when it is ready and the person chose "Open F1R3Games when ready" at first run, open `http://localhost:40700/` in the default browser. Only at first run; later starts open nothing.

## 10.5 Health

The portal is *ready* when `GET /api/ready` (F5) answers 200, meaning: the node is reachable, the registered portal environment version equals the configured one, and every game recorded as registered is listed with the recorded hashes. Until F5, the supervisor checks `GET /api/env` (the URI equals the recorded one) and `GET /api/games` (each recorded game present). F1R3Games is *Degraded* when the portal is ready but the observer trails or a game's hashes differ from the bundle's (an upgrade is pending), and *Waiting* whenever the shard is not Running.

## 10.6 What the person does in the browser

1. `http://localhost:40700/`: the feed. "Sign on" creates a keystore protected by a passphrase, and optionally a passkey, or imports a key file (the portable format F1R3Gaze and F1R3Sky

use, so the person's F1R3Gaze wallet key can be brought in). The page pins the portal environment URI and asks the faucet to fund the new address with 100 F1R3 (Decision 14).

2. /games: F1R3Pix and F1R3Beat, from `games.register`.
3. /games/:id/launch: a new instance, unlisted by default, with the game's configuration (F1R3Pix: capacity 61, random seating; F1R3Beat: 4/4, two bars, sixteenths, capacity 32, 100 bpm). The launch sets an allowance so moves are signed without a prompt; payments are always prompted.
4. /instances/:id: the lobby; invite (a link at localhost, useful to other browsers and other accounts on this Mac), start, play in the framed client.
5. /games/:id/gallery: plays by kind and day, each previewed by the game's renderer from the game's own origin; /plays/:id opens one.

ign1t10n touches none of this; it guarantees only that the portal, its origins, the environments and the registrations are there.

## 10.7 The F1R3Beat breeder (optional)

F1R3Beat's patterns breed in epochs run by a breeder key that F1R3Beat's environment key has named (F1R3Beat design v2 §10). When "Run the F1R3Beat breeder daily" is on (Decision 15), ign1t10n once signs `setBreeder` with F1R3Beat's game key, launches a public nursery instance with the breeder's key, records its id, and then runs the breeder job (`beat-epoch`, F6) once a day while the shard runs, skipping an epoch when nothing has been published since the last.

## 10.8 Upgrade

At start, the supervisor compares `Resources/f1r3games/games.toml` with what `shard.toml` records:

- a client-only change: nothing to do; the new bundle's files are served;
- a game's template hashes or environment version changed: run G5 for that game at the recorded version +1 and G6, *after consent*, since its live instances are replaced (Finding 17); until then the game stays registered at the old hashes and F1R3Games is Degraded ("F1R3Beat update waiting");
- the portal environment's version rose: the same, with a dialog saying that profiles, instances, plays, invitations and sponsorships on the local shard will be discarded; browser keystores are unaffected (the URI is unchanged).

## 10.9 The F1R3Gaze rendition

The shell's core/ layer is framework-free and talks to the world through eight capabilities, so that a f1r3lang page in F1R3Gaze can do the same (`docs/GAZE-MAPPING.md`). When F1R3Gaze gains those capabilities, ign1t10n will point F1R3Gaze at the portal's site (`f1r3://f1r3fly/games`) as well; nothing on the chain changes. Until then the menu opens the portal in the default browser and F1R3Gaze in its own window, and the two are independent.

## 11 User interface

### 11.1 Menu bar

A monochrome template image of the F1R3FLY.io firefly with a status dot: none for Running, amber for Starting, Resizing, Degraded and F1R3Games Installing, red for Failed, hollow for Stopped.

| Item                                      | Action                                                               |
|-------------------------------------------|----------------------------------------------------------------------|
| <i>Local shard: Running, 2 validators</i> | submenu: height, per-node readiness, observer lag, disk              |
| <i>F1R3Games: Running, 2 games</i>        | submenu: portal address, each game and its origin, last job, breeder |
| Open F1R3Games                            | open <code>http://localhost:40700/</code> in the default browser     |
| Open F1R3Gaze                             | open <code>-b io.f1r3fly.f1r3gaze</code>                             |
| Start / Stop shard                        | through the control socket                                           |
| Configure... (Command-comma)              | the configuration panel                                              |
| Fund a wallet...                          | faucet transfer to any address (a F1R3Gaze or portal key)            |
| Copy endpoints                            | validators, observer, shard id, Embers, portal                       |
| Show logs                                 | reveals <code>~/Library/Logs/F1R3FLY/ign1t10n</code>                 |
| Reset shard... / Uninstall...             | §14                                                                  |
| Quit ign1t10n                             | quits the menu bar only                                              |

### 11.2 The configuration panel

The **Shard** and **General** panes are version 0.3's (validator stepper with footprint and measured tolerance, node table, Apply sheet; run at login, use the local shard in F1R3Gaze, ports, genesis amounts, Embers on/off). A third pane is new.

#### F1R3Games.

- **Install F1R3Games** on/off (`ctl games on|off`). Off stops the portal and frees its ports; it leaves the environments and registrations on the chain and the keys in the Keychain, so turning it on again needs no deploy.
- **Address:** `http://localhost:40700`, read-only, with “Open” and “Copy”; “Move to a new address...” behind the dialog of §8.5.
- **Games:** one row per game: origin, environment version, registered or not, client bundled or not, pending update with “Update...”.
- **Faucet amount** for new portal keys (default 100 F1R3).
- **Run the F1R3Beat breeder daily** (default off) with the last epoch.
- **Reinstall** (re-runs G3–G6; nothing is deployed that is already there) and **Rotate F1R3Games keys...** (danger: new environment URIs; every browser keystore must be cleared, every game re-registered).

### 11.3 First-run window

F1R3FLY.io brand on a white surface (white-background logo; yellow and sky only as rules and progress), listing S0–S9 and then G1–G7, each with a spinner, check or error, and the “Advanced” disclosure of §7. It closes with a summary: shard id, validators, the funded F1R3Gaze wallet, the portal address with an “Open F1R3Games” button, and where the logs are.

## 12 Required changes

### 12.1 F1R3Gaze

G1 (porcelain wallet commands), G2 (balances without Embers), G3 (`settings.d`), G4 (a validator list, one drawn per deploy) and G5 (a shard health page) are version 0.3's and `IMPLEMENTATION.md`'s, unchanged.

### 12.2 F1R3Node-Rust

N1 (bind addresses for transport and discovery), N2 (vendored OpenSSL and a macOS CI build), N3 (the loopback shard test with grow and shrink), N4 (storage-format number) and N5 (`api-server.reject-foreign-origin`) are unchanged (`docs/node-work-packages.md`). N5 does not conflict with F1R3Games (Finding 12).

### 12.3 F1R3Games

**Work package 1** (F1: keys from the environment). `f1r3games-service` reads `F1R3GAMES_SERVICE_KEY`, `F1R3GAMES_ENV_KEY`, `F1R3GAMES_TOKEN_SECRET`, `F1R3GAMES_GAME_KEY_<ID>`, `F1R3GAMES_COOP_KEY` and `F1R3GAMES_BREEDER_KEY` (hex) when set, before the configured files, and never logs them. Estimate: 0.25 engineer-week.

**Work package 2** (F2: game origins served by the service). A repeated `[[origins]]` table: `id`, `listen` (a list), `dir`; each serves only `/<id>/` from `dir` (index fallback within it, no directory listing) with the headers of §10.2, on its own listeners in the same process. Estimate: 1 engineer-week.

**Work package 3** (F3: headless registration and per-game entries). `games-manifests --entry ID=URL` (repeatable; `--entry-base` stays the default) and `f1r3games-service register-games FILE [--only ID]`, which signs `games.register` with `F1R3GAMES_COOP_KEY`, refuses when its address is not the environment's `coop_address`, skips a game already registered with the same manifest, and waits for finalisation. Estimate: 0.5 engineer-week.

**Work package 4** (F4: dual-stack listeners and a Host allow-list). `listen` accepts a list; the default becomes `["127.0.0.1:P", "[::1]:P"]`. A configured `public_host` (`localhost:40700`) is the only Host served; loopback literals are redirected to it; anything else gets 421. The same for each `[[origins]]` entry. Estimate: 0.5 engineer-week.

**Work package 5** (F5: readiness and status). `GET /api/ready` (§10.5) and `f1r3games-service status --json`: the portal environment's registered version, each game environment's, and each registered manifest's `id`, `entry` and template hashes. Estimate: 0.5 engineer-week.

**Work package 6** (F6: a headless breeder (optional)). `f1r3games-service beat-set-breeder` and `beat-epoch --nursery ID`, signing with the keys of F1, computing the epoch as `f1r3games beat epoch` does. Estimate: 0.5 engineer-week.

**Work package 7** (F7: several validators (optional)). `validator_urls`, one drawn per deploy, with a retry on another when one refuses, so the portal survives a validator's restart and needs no restart when one withdraws. Estimate: 0.25 engineer-week.

## 13 Security model

| Threat                           | Protection                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reuse of public development keys | All keys generated on the Mac; the release pipeline fails if any public key from <code>docker/.env.example</code> or any address from F1R3Games's <code>examples/local-shard/</code> appears in an artifact or in <code>shard.toml</code> .                                                                                        |
| Another user on the Mac          | State directory 0700; no key in any argument vector; passwords and F1R3Games keys only in the Keychain and in the owning process's environment; control socket checks the peer uid. Loopback is shared between accounts: another user can open the portal and play, as any player can, and use its faucet; the faucet is the cost. |
| Hosts on the same network        | Every listener on loopback (N1, <code>api-server.host</code> , F4).                                                                                                                                                                                                                                                                |
| Pages in other browsers or tabs  | Cannot use a node (N5). Can reach the portal's API only cross-origin: no CORS origin is configured, so responses are unreadable and JSON POSTs are refused at preflight. DNS rebinding is refused by the Host allow-list (F4).                                                                                                     |
| A malicious or buggy game        | Own origin and sandbox; the host answers only that frame and that origin; the wallet signs only the game's registered play templates within the launch allowance, refuses fund-moving templates from a game, and prompts for payments (F1R3Games design).                                                                          |
| Theft of a browser key-store     | Encrypted under the passphrase (PBKDF2) or the passkey's PRF output; it never leaves the browser profile; <code>ign1t10n</code> never reads it.                                                                                                                                                                                    |
| The local Cooperative key        | Governs only the registry of this local shard; it is never given to the portal process.                                                                                                                                                                                                                                            |
| Tampered artifact                | Developer ID signatures on every Mach-O and the archive; notarisation; hardened runtime with no entitlements; clients served from inside the signed bundle; GPG-signed SHA256SUMS.                                                                                                                                                 |
| Malware running as the user      | Not defended.                                                                                                                                                                                                                                                                                                                      |

## 14 Upgrade, reset and uninstall

### 14.1 Upgrade

A newer archive replaces both bundles. The supervisor compares the node's storage format (N4) with the recorded one and restarts every node on its data, or offers a new shard; then it runs §10.8 for F1R3Games.

### 14.2 Reset

"Reset shard..." archives the chain, node keys and configurations and re-provisions from S1 with the chosen  $N_0$ . `games.secrets` is kept, so the environment URIs are the same; G3–G6 run again on the new chain, and the portal's origin is the same. The dialog says that every deploy, registry entry, balance, profile, instance, play and gallery on the local shard will be gone; that F1R3Gaze wallets and browser keystores keep their keys and can sign on again at once; and that the faucet will fund them again. A checkbox "Also rotate the F1R3Games keys" (off) does what the panel's rotation does.

## 14.3 Uninstall

Stop everything; unload and remove the agent; after a second confirmation remove the state directory (optionally keeping archive/), the logs and every Keychain item of service `io.f1r3fly.ign1t10n` (including `games.secrets`); remove F1R3Gaze's managed block or drop-in; move `ign1t10n.app` to the Trash; `pkgutil --forget io.f1r3fly.ign1t10n.pkg`. F1R3Gaze and its wallets are left alone. `ign1t10n` cannot reach browser storage: the dialog says that keystores remain in each browser under `localhost:40700` until the person clears that site's data, and offers to open the portal's Wallet page first so keys can be exported.

## 15 Build and release pipeline

`versions.toml` gains a section, read by the release workflow and copied into the bundle:

```
[f1r3games]
repository = "https://github.com/F1R3FLY-io/F1R3Games"
revision   = "57a3b2796108f90dd4689abf33659a34c351d2ef" # branch f1r3beat; main once
merged
toolchain  = "stable" # Rust >= 1.85
node       = "22"     # npm ci for web/ and the
clients    clients
clients    = ["f1r3pix", "f1r3beat"] # Decision 11
# F1R3Games work packages this revision contains (F1..F7).
has = []
```

| Job         | Runner   | Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| node        | macos-14 | F1R3Node-Rust at the pinned revision, <code>--features crypto/vendored-openssl, arm64; otool -L gate</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| embers      | macos-14 | Embers at the pinned revision, arm64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| gaze        | ubuntu   | download and verify the pinned F1R3Gaze release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| games (new) | macos-14 | F1R3Games at the pinned revision: <code>cargo test</code> for the workspace and <code>rhoint</code> ; <code>cargo build --release -p f1r3games-service -p f1r3games-cli, arm64; web/</code> : <code>npm ci</code> , build the wallet to WebAssembly ( <code>wasm32-unknown-unknown, wasm-bindgen 0.2.129</code> ), <code>npm test</code> , <code>npm run build</code> ; each listed client: <code>npm ci</code> , <code>npm test</code> , <code>npm run build</code> ; write <code>games.toml</code> (environment versions and template hashes from <code>games-manifests</code> ); <code>otool -L gate</code> |
| ign1t10n    | macos-14 | build, unit and end-to-end tests (the stand-in node gains a stand-in portal)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| assemble    | macos-14 | <code>build.sh</code> with the F1R3Games artifacts; sign inside out ( <code>Helpers/*</code> , <code>MacOS/ign1t10n</code> , the app); <code>pkgbuild</code> , <code>productbuild</code> , sign, notarise, staple                                                                                                                                                                                                                                                                                                                                                                                              |
| smoke       | macos-14 | headless provisioning with F1R3Games on; A2–A6, A11, A12, A15, A17, A23; A16 in headless Chromium and WebKit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| publish     | ubuntu   | SHA256SUMS, GPG signature; draft release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## 16 Testing and acceptance

A1–A14 are version 0.3's: clean install; round trip through F1R3Gaze; loopback only; no foreign libraries; no shipped secrets; persistence across reboot; port conflict with a Docker shard; crash

recovery; upgrade; uninstall; grow in place; shrink in place; new shard at the maximum; bounds and interruption. A1 now also requires F1R3Games Running within seven minutes. New:

#### **A15 F1R3Games installed.**

After A1, `ctl games status --json` reports the portal ready, six environments at their bundled versions, and F1R3Pix and F1R3Beat registered with entries at `localhost:40701` and `:40702`.

#### **A16 Playing in a stock browser.**

Driven by Playwright in Chromium, WebKit and Firefox: sign on with a passphrase; the faucet funds the key; launch an F1R3Pix instance; seat, paint, publish a canvas; the gallery lists it and the renderer draws it; then the same for F1R3Beat with a pattern. Repeat sign-on with a passkey where the browser supports the PRF extension.

#### **A17 Origins.**

`http://127.0.0.1:40700/` redirects to `localhost`; a request with `Host: evil.example` gets 421; the portal refuses to frame a game whose entry is its own origin; game responses carry `frame-ancestors`; one game's origin does not serve another's files.

#### **A18 Dual stack.**

With a decoy listener on `[::1]:40700` before provisioning, G2 allocates another decade. With one started after provisioning, F1R3Games goes Failed naming the holder and does not move.

#### **A19 Reset keeps identities.**

A keystore created before "Reset shard" signs on afterwards without a pin error; the environment URIs are unchanged; G3–G6 ran.

#### **A20 Upgrades.**

A bundle whose only change is a client deploys nothing; one with a changed F1R3Beat template asks consent, installs version +1 and re-registers; F1R3Pix is untouched.

#### **A21 Evacuation.**

Shrinking away the portal's validator moves the portal first; a move sent during the shrink finalises.

#### **A22 Off and on.**

`ctl games off` frees ports 40700–40702; on restarts with no deploy and the same origin.

#### **A23 Loopback.**

`lssof` lists `f1r3games-service` only on `127.0.0.1` and `::1`.

## 17 Work packages and schedule

| WP     | Deliverable                                                                                                                           | Est. | Status                |
|--------|---------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------|
| N1–N5  | node: bind addresses, macOS build, loopback shard test, storage format, origin guard                                                  | 4.5  | open                  |
| G1–G5  | F1R3Gaze: porcelain, balances, <code>settings.d</code> , validator list, health page                                                  | 2.5  | open                  |
| H1–H6  | ign1t10n: provisioning, supervision, UI, resize, panel, lifecycle                                                                     | 11   | first working version |
| R1, T1 | release workflow; acceptance and measurements                                                                                         | 3.5  | partly                |
| F1–F5  | F1R3Games: keys from environment, game origins, headless registration, dual stack and Host, readiness                                 | 2.75 | new                   |
| F6, F7 | F1R3Games: headless breeder; validator list (optional)                                                                                | 0.75 | new                   |
| H7     | ign1t10n: manifest schema 4, <code>games.secrets</code> , stages G1–G7, the portal role, jobs, health, evacuation, upgrade comparison | 2.5  | new                   |
| H8     | ign1t10n: menu items, F1R3Games pane, first-run additions, opening the browser                                                        | 1    | new                   |
| H9     | ign1t10n: reset, rotation, uninstall and upgrade dialogs for F1R3Games                                                                | 1    | new                   |
| R2     | release: the games job, bundle layout, signing, smoke additions                                                                       | 1    | new                   |
| T2     | A15–A23; browser matrix; footprint and on-chain growth of plays                                                                       | 1.5  | new                   |

The new work is about 10.5 engineer-weeks. F2, F3 and F5 are on its critical path: H7 needs game origins to register entries, headless registration to finish G6, and readiness to supervise. F1 and F4 can follow (H7 has an interim for F1). With one engineer on F1R3Games (F1–F7) and one on ign1t10n (H7–H9, R2), and T2 shared, the component fits in five to six calendar weeks, alongside the open version 0.3 work.

## 18 Obligations and decisions

### 18.1 Obligations

Version 0.3’s obligations stand: the loopback shard works (N3); resizing works (N3); footprint (T1); measured fault tolerance (T1); PEM compatibility (discharged at 46faa56). Added:

**Obligation 1** (The environments behave on a node). Before H7 is merged, T2 must show, on an ign1t10n shard at the pinned node, the portal environment and the F1R3Pix and F1R3Beat environments doing what their designs say, including the parts F1R3Games’s `IMPLEMENTATION.md` lists as unverified (the `TreeHashMap` handle, sponsorship escrow, `/\ Int` and `/\ String` patterns, phlo per move), and record phlo per move and per read so the allowance and capacity defaults rest on measurements.

**Obligation 2** (Browsers). T2 must establish, for current Safari, Chrome and Firefox on macOS 13–26: that `http://localhost` is a secure context for WebAuthn, IndexedDB and WebAssembly; which support the PRF extension the passkey unlock needs (the passphrase works in all); and that the sandboxed game frames with `allow-same-origin` exchange messages with the portal. The first-run summary names any browser found wanting.

**Obligation 3** (Origin stability). The end-to-end test must show that no path through start, port conflict, resize, reset, upgrade or F1R3Games off/on changes a recorded F1R3Games port.

**Obligation 4** (Embers compatibility). Embers 28e50d4’s contracts do not parse at node fce422a. F1R3Games does not depend on Embers, but the release must not ship a pair that fails: pin a compatible Embers or default Embers off, and make the smoke job deploy its contracts.

## 18.2 Resolved on 30 September 2026

| # | Decision                            | Resolution                                                                                                                                                                         |
|---|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Fold the bootstrap into validator 1 | No: keep the separate bootstrap for version 1.                                                                                                                                     |
| 2 | Intel                               | Apple silicon only.                                                                                                                                                                |
| 3 | Minimum macOS                       | 13.0.                                                                                                                                                                              |
| 4 | Genesis amounts                     | 10 000 000 F1R3 to the browser’s wallet; validators 1 000 F1R3 fees and 1 000 stake; the rest of 500 000 000 to the faucet.                                                        |
| 5 | Where deploys go                    | A randomly selected active validator, redrawn at each start and after each resize; with G4, the whole list.                                                                        |
| 6 | Timings                             | Tuned for a laptop (profile laptop-v1: casper loop 1.0 s at $N = 2$ rising to 3.0 s at $N = 10$ ; heartbeat on; maximum LFB age 15 s, or 30 s above $N = 4$ ); recalibrated by T1. |
| 7 | Foreign Origin                      | The node refuses it (N5).                                                                                                                                                          |
| 8 | Embers                              | An installation option, default on.                                                                                                                                                |

## 18.3 Open decisions

**Open decision 9** (The F1R3Games port block). 40700–40709, beside ign1t10n’s other blocks and apart from a developer’s 8640, 5173 and 5174 (proposed); or the service’s own default, 8640, which would collide with a development portal.

**Open decision 10** (Who is the Cooperative on the local shard). A key ign1t10n generates and holds (proposed): registration then needs nothing from the person, and the key governs nothing outside this shard. Or the person’s own key, imported, so that the same identity governs every shard they run.

**Open decision 11** (Which games to register). Install all five environments; register only games whose web client is bundled, F1R3Pix and F1R3Beat (proposed), adding the others as clients land. Or register all five now, which lists three games the browser cannot play.

**Open decision 12** (One origin per game). Per-game origins (proposed), as the portal’s isolation model intends; or one shared games origin, which is simpler but lets games read one another’s storage.

**Open decision 13** (Opening the browser). At the end of first run, open the portal in the default browser as well as F1R3Gaze (proposed); or show only the “Open F1R3Games” button.

**Open decision 14** (Faucet amount). 100 F1R3 per new portal key on the local shard (proposed), enough for allowances and F1R3Pix and F1R3Beat payments between players; or the service default of 1 F1R3.

**Open decision 15** (The F1R3Beat breeder). Off by default, a switch in the panel (proposed), since a single person’s local shard produces little engagement to select on; or on, daily.

**Open decision 16** (Ship the CLI). Bundle f1r3games in Helpers with `ctl games env` printing how to point it at the local portal (proposed); it is how a developer scripts plays and runs `beat verify`.

**Open decision 17** (F1R3Games by default). On by default in the first-run window, like Embers (proposed).

## A Configuration templates

### A.1 Node configuration

`conf/common.conf`, `bootstrap.conf`, `validator-{k}.conf` and `observer.conf` are version 0.3's, with the `laptop-v1` timing values in place of `docker/conf/default.conf`'s and `logging.sink = "stdout"`:

```
# conf/common.conf (excerpt): rendered by ignit10n; values for N validators
protocol-server { network-id = "{{NETWORK_ID}}", bind-address = "127.0.0.1", # N1
                  allow-private-addresses = true, no-upnp = true }
protocol-client.network-id = "{{NETWORK_ID}}"
peers-discovery { bind-address = "127.0.0.1", lookup-interval = 60 seconds }
api-server { host = "127.0.0.1", reject-foreign-origin = true } # N5
casper {
  shard-name = "root"
  fault-tolerance-threshold = 0.1
  casper-loop-interval = {{LOOP_MS}} ms # 1000 at N = 2 .. 3000 at N = 10
  heartbeat { enabled = true, max-lfb-age = {{LFB_AGE}} seconds }
  genesis-ceremony { required-signatures = {{max(N0-1,1)}},
                    approve-interval = 10 seconds, approve-duration = 1 minutes }
  genesis-block-data { bonds-file = "{{STATE}}/genesis/bonds.txt",
                      wallets-file = "{{STATE}}/genesis/wallets.txt",
                      number-of-active-validators = 10,
                      epoch-length = 10, quarantine-length = 10 }
}
logging { format = "json", sink = "stdout" }
```

### A.2 games/f1r3games.toml

```
# Rendered by ignit10n at G4 and at every start (the validator is redrawn).
listen = ["127.0.0.1:40700", ":::1:40700"] # F4; one address until it
lands
public_host = "localhost:40700" # F4
shard_id = "root"
validator_url = "http://127.0.0.1:40420" # a randomly drawn active
validator
observer_url = "http://127.0.0.1:40453"
coop_address = "{{LOCAL_COOPERATIVE_ADDRESS}}"
env_version = 1 # from Resources/f1r3games/
games.toml
phlo_price = 1
default_phlo_limit = 500000
max_phlo_limit = 50000000
```

```

env_phlo_limit      = 50000000
deploy_ttl_ms       = 600000
token_ttl_secs      = 3600
bootstrap_env       = false           # installation is a job (G4), not a side effect of
    serve
cors_origins        = []              # the shell is same-origin
portal_base_url     = "http://localhost:40700"
static_dir          = "/Applications/ign1t10n.app/Contents/Resources/f1r3games/portal"
# keys come from the environment (F1); until then, *_file paths in a 0700 run directory

[faucet]
enabled = true
amount  = 100000000000               # 100 F1R3 (Decision 14)

[[origins]]                # F2
id      = "f1r3pix"
listen  = ["127.0.0.1:40701", "::1:40701"]
public_host = "localhost:40701"
dir     = "/Applications/ign1t10n.app/Contents/Resources/f1r3games/games"

[[origins]]
id      = "f1r3beat"
listen  = ["127.0.0.1:40702", "::1:40702"]
public_host = "localhost:40702"
dir     = "/Applications/ign1t10n.app/Contents/Resources/f1r3games/games"

```

## B The manifest

Schema 4 adds [options].games, [options].games\_open\_at\_first\_run and the [games] tables to schema 3; an absent [games] reads as F1R3Games never installed, so a schema-3 manifest upgrades in place.

```

# ~/Library/Application Support/F1R3FLY/ign1t10n/shard.toml (excerpt)
schema = 4

[options]
embers = true
gaze_integration = true
deploy_target = "random"
timing = "laptop-v1"
games = true
games_open_at_first_run = true

[games]
revision = "57a3b27"           # the F1R3Games revision last installed
portal_port = 40700
validator_slot = 2             # drawn at each start
service_address = "1111...Xs"
coop_address = "1111...Cp"
breeder_address = "1111...Br"
env_uri = "rho:id:...g7"
env_version = 1

```

```

faucet_f1r3 = 100
breeder = false

[[games.game]]
id = "f1r3pix"
port = 40701
env_uri = "rho:id:i77c...gtgn"
env_version = 2
client = true
registered = true
manifest_sha256 = "5e0c...91"

[[games.game]]
id = "f1r3beat"
port = 40702
env_uri = "rho:id:...q4"
env_version = 1
client = true
registered = true
manifest_sha256 = "a3b1...0d"

[[games.game]]
id = "f1r3ink"
env_uri = "rho:id:...k2"
env_version = 1
client = false
registered = false

[games.stages]
secrets = true
ports = true
funded = true
portal_env = true
game_envs = true
registered = true
opened = true

[games.job]
kind = "register"
game = "f1r3beat"
started = 2026-10-14T09:20:03Z
deploy = "3045...aa"

```

*# environment installed, not registered*

*# G1..G7, cleared from G3 at reset*

*# present only while a job runs*